

The IP₃ receptor's binding core is not more constrained than its pore, and the answer depends on where the pore ends

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Abstract

The IP₃-binding core is the one part of the IP₃ receptor that the ryanodine receptors, which share its fold and pore, do not share functionally. We asked whether that core is under different constraint from the pore it gates. Paired inside each orthologue, across 246–262 vertebrate orthologues per paralogue, the pore module is more conserved than the ligand core in *ITPR1* and *ITPR3* and level in *ITPR2*. All three paralogues reverse when the luminal loop is counted as pore, so the comparison is only interpretable with that boundary declared. Measured as distance to the ligand in six independent IP₃-bound structures, the ten contact residues are more constrained than the rest of the core and no more constrained than the rest of a pocket about 15 Å across, and per-site substitution rates show the same shape. Across 3,527 eukaryotic reference proteomes, 64 carry an IP₃ receptor and no phosphoinositide-specific phospholipase C. Matched on divergence, their ligand core is not relaxed, and the null is bounded by a ryanodine receptor control that the matched test detects with essentially full power.

Introduction

Inositol 1,4,5-trisphosphate (IP₃) releases calcium from the endoplasmic reticulum by opening the IP₃ receptor, a tetrameric channel of about 2,700 residues per subunit [1, 2]. Vertebrates carry three paralogues, *ITPR1*, *ITPR2* and *ITPR3*. The ligand binds at the N-terminus, in a core formed by a β -trefoil and an armadillo-like domain [3, 4], and the signal travels some two thousand residues to the pore at the C-terminus [5, 6, 7].

The ryanodine receptors are the IP₃ receptor's sister family. They share the receptor's architecture, its pore and every domain used to diagnose it [8, 9], and they do not bind IP₃; their gating is organised around other ligands [10]. The ligand core is therefore the one module whose function separates the two families, which makes it the natural place to expect the tightest constraint in the receptor. That expectation has not been measured, for three reasons this paper addresses.

First, neither module is a database domain. Pfam names the ligand region after the families it identifies rather than after the ligand, and its pore domain includes a luminal loop that is the least conserved sequence in the receptor. A comparison drawn on database spans therefore compares something other than the ligand core against something other than the pore. We define both modules by measurement, and we run every test under a second, conventional definition of each.

Second, the ligand site is usually described as a set of contact residues. A contact label discards what a structure measures, the distance of every residue from the ligand, and it cannot say whether selection holds the contacts or the neighbourhood around them. We measure the site as a distance, all-atom, in six independent IP₃-bound depositions.

Third, if the core is constrained because it binds IP₃, lineages that no longer make IP₃ should relax it. Which lineages those are is usually taken from reading. We derive them by sweeping every eukaryotic reference proteome for the enzyme that makes IP₃, and we test the prediction against a positive control, the ryanodine receptors, whose ligand-free core shows what relaxation looks like on the same instrument.

The question this paper answers is whether the IP₃-binding core is under different constraint from the pore it gates (Fig. 1). The per-residue conservation layers it stands on are built and described in companion paper 5. The comparison, its controls and its scope are all measured here.

Results

Both modules are defined by measurement and checked against what they must not contain

We defined the primary ligand core as the minimal contiguous span containing every IP₃ contact measured on the 6DQN structure [6], residues 265–569 in *ITPR1*, and the primary pore module as the Pfam channel domain less the geometrically located luminal loop, residues 2363–2608 in *ITPR1* with the loop removed (Fig. 1a). Each definition was checked against residues it must not hold: the core holds all ten contacts and no filter or gate residue, and the pore both filter and both gate residues and no contact. A failing definition stops the analysis rather than being written. The sensitivity definitions are the published IP₃-binding core [3], transferred to each paralogue, and the channel domain as InterPro draws it, loop included. Both modules are laid out at residue resolution in all three paralogues in Supplementary Fig. 1.

The pore is more conserved than the ligand core in two paralogues and level in the third

Every comparison is paired inside one sequence: each orthologue contributes one core identity and one pore identity to its own paralogue's reference, and only their difference enters the test, so a generally divergent sequence cannot favour either module. The tests ran on 260 *ITPR1*, 246 *ITPR2* and 262 *ITPR3* orthologues.

The pore module is ahead in *ITPR1* (mean difference -0.0239; 223 orthologues favour the pore against 32 the core; $q = 4 \times 10^{-34}$) and in *ITPR3* (-0.0199; 218 against 42; $q = 2 \times 10^{-28}$). *ITPR2* shows no difference ($q = 0.13$) (Fig. 1b). The module that binds the ligand, and whose function separates this family from its sister, is the less conserved of the two across vertebrate orthologue sets.

The answer reverses in all three paralogues when the luminal loop counts as pore

Leaving the luminal loop inside the pore module, as the database draws the domain and as a comparison would be made by default, reverses the sign in every paralogue. The ligand core then leads by 0.055 in *ITPR1*, 0.0445 in *ITPR2* and 0.0558 in *ITPR3*, each at $q < 10^{-37}$ (Fig. 1b). The loop's fifty residues are the least conserved stretch of the receptor, and they sit inside the same domain as the gate. Neither answer is wrong about its own module. They are answers about different modules, and a comparison that does not state where the pore ends cannot be read.

Swapping in the published binding core moves one paralogue: with the loop excluded, *ITPR2* tips from level to a slight lead for the core (0.0049, not significant), while *ITPR1* and *ITPR3* keep the pore ahead. The luminal-loop boundary decides the sign everywhere, and the core boundary decides it only where the two modules were already level.

One further disagreement is reported rather than dropped. Scored column by column with the Jensen–Shannon divergence [11], the two modules do not differ, while the composition-free share of the modal residue agrees with the paired test. The divergence is measured against a background amino-acid table, so a transmembrane module scores lower than a soluble one at equal conservation; the column-level null is a property of that metric.

Six independent structures place the same ligand pocket

We measured every residue within 15 Å of IP₃, using all atoms rather than C α , in six IP₃-bound human IP₃R3 depositions from several groups and gating states [12, 6, 7, 13]. Recovery of the ten published contacts in the 6DQN structure was required before anything downstream ran, and all six depositions recover all ten. In two of them, 8TKG and 8TKH, that depends on a modelled hydrogen: by heavy atoms alone, the rule the ten were defined with, Arg503 sits 4.78 and 4.83 Å from IP₃, just outside the cutoff. They also agree on two more residues, Ala276 and Arg411, each within 4.5 Å in four of the six depositions and outside it in 6DQN. The pocket so defined holds 125 residues, binned into a contact shell of twelve residues (≤ 4.5 Å) and three outer shells out to 15 Å, the outermost holding 59 (Fig. 2a).

Selection holds a pocket rather than a contact set

The ten contacts are more constrained than the rest of the ligand core in all three paralogues (mean deep-layer divergence 0.813 against 0.744 in *ITPR1*; $q = 0.048, 0.017$ and 0.041), with the permutation resampling which positions carry the contact label while the constraint values stay where they are. Against every other residue within 15 Å of the ligand they are more constrained in none ($q \geq 0.12$).

Every shell out to 15 Å sits above the whole-protein mean (0.7366 in *ITPR1*), and the step a contact-driven model predicts at 4.5 Å is absent (Fig. 2b). Conservation falls with distance across the pocket, strongly in *ITPR2* ($\rho = -0.436$) and weakly in the other two, where the trend does not reach significance. What selection holds at the ligand site is a neighbourhood about 15 Å across, which a contact label cannot see.

Per-site substitution rates reproduce the pocket but not the module ranking

Column conservation and a substitution rate on a tree are both called constraint and are different measurements, so we asked the same two questions of per-site rates estimated by FEL [14] on the vertebrate codon alignments (Fig. 3). Every contact site in every paralogue is under detectable purifying selection. That share falls to 0.875, 0.75 and 0.825 in the third shell of *ITPR1*, *ITPR2* and *ITPR3*. The fall is not monotone, since the outermost shell sits slightly above the third (0.8966 in *ITPR1*), so the data show a step down from the first two shells onto a floor rather than a gradient.

The module ranking does not replicate on this axis, and in *ITPR1* it points the other way: the pore's mean nonsynonymous rate (0.0446) exceeds the core's (0.0251). In *ITPR3* the core evolves faster ($q = 0.058$), and *ITPR2* shows nothing. The rate axis uses about a quarter as many sequences as the identity axis and a rate that is zero at most sites, so we read the module result off the identity axis and report the disagreement beside it.

The enzyme that makes IP₃ is missing from 64 proteomes that carry the receptor

We swept both halves of the phosphoinositide-specific phospholipase C (PI-PLC) catalytic barrel over 3,527 eukaryotic reference proteomes and counted a proteome as carrying the enzyme only when one protein held both halves. The 763 vertebrate proteomes are the search's positive control: 760 carry a PI-PLC, and none carries a receptor without one.

Across the whole sweep, 64 proteomes carry an IP₃ receptor and no PI-PLC (Fig. 4a): 22 fungal, 28 from other protists, 10 non-vertebrate animal and 4 green-plant proteomes. They concentrate in the oomycetes and

the early-diverging fungi. Every one of them carries a full-length receptor, and a gene set complete enough to hold a 2,700-residue multi-exon gene is complete enough to hold a phospholipase.

Matched on divergence, losing the enzyme does not relax the ligand core

The representative alignment holds only two of these lineages, which is too few to test, although the ryanodine receptor control already fires there (shift -0.0996 on six tips). We therefore aligned all 662 non-vertebrate receptors from reference proteomes to the human reference through the same module residues; 486 resolve both modules and enter.

Pooled, the 36 enzyme-absent receptors show a wider core-minus-pore gap than the 450 others ($p = 9.7 \times 10^{-6}$). The pooled comparison is confounded by clade: enzyme-absent proteomes sit at a median pore identity of 0.3582 against 0.5886 for the rest, and the paired difference itself correlates with divergence (Fig. 4b). Inside a phylum, where the clade is held constant, the comparison is possible in only three strata. All 15 early-diverging fungal (Mucoromycota) receptors lack the enzyme, so that phylum has no internal control.

We therefore matched each enzyme-absent receptor to up to three enzyme-carrying receptors within 0.03 pore identity of its own. All 36 matched. The median within-pair difference is -0.0064 (95 % CI -0.0161 to 0.0152), with 19 pairs on one side and 17 on the other (sign test $p = 0.87$).

The null is bounded. Measured through the same pairwise instrument at the same divergence, the ryanodine receptors shift the paired difference by -0.0624. The matched test detects a shift of 0.05 with power 0.99 and one of 0.03 with power 0.59 (Fig. 4c), and its interval excludes shifts larger than about 0.016. Losing the enzyme that makes IP₃ does not relax the receptor's IP₃-binding core, and a ligand-free core at this divergence would have been detected.

Discussion

The IP₃-binding core is not the most constrained part of the IP₃ receptor. Paired per orthologue across roughly 250 sequences per paralogue, the pore module leads in *ITPR1* and *ITPR3* and the two are level in *ITPR2*. The module that gives the family its name and its diagnostic signature is the less conserved of the two. Naming a family after its diagnostic domain is a statement about what that domain identifies, and it does not predict what selection holds most tightly.

The answer depends on one boundary. With the luminal loop counted as pore, all three paralogues reverse and the core leads by about five identity points. The loop is solvent-facing, poorly resolved in the cryo-EM maps [6, 7] and the least conserved sequence in the receptor, and it lies inside the same database domain as the gate. Any comparison of ligand-binding and pore regions in this family, and plausibly in its sister family, has to declare that boundary to be interpretable.

At the ligand site itself, selection holds a pocket and not a contact set. The ten residues that touch IP₃ are special against the rest of the binding core and not against the other residues within 15 Å of the ligand, and constraint shows no step at the contact radius. The same shape appears on an independent axis, the per-site substitution rate on a vertebrate codon tree. The structural studies that define the binding core describe a pocket lined by basic residues from two domains [3, 4]; our result says that evolution treats that whole pocket, rather than its phosphate-coordinating residues alone, as the conserved unit.

Losing the enzyme that makes IP₃ does not relax the core. Sixty-four eukaryotic proteomes carry the receptor and no phosphoinositide-specific phospholipase C, most of them oomycetes and early-diverging fungi. The comparison that looks positive when pooled is a comparison of distant against near, and it vanishes when each record is matched on divergence. The null is bounded by the ryanodine receptors, which have the same pore and no IP₃ site and move the paired difference roughly ten times further than the matched interval allows.

Four limits apply. First, whether the pocket's constraint concerns IP₃ at all is not settled: every residue within 15 Å of the ligand is also inside the fold that holds it, and sequence cannot separate ligand binding from domain packing. Mutational or binding-affinity data could. Second, a PI-PLC search looks for one enzyme family. An absence says the canonical route to IP₃ is missing, not that the receptor lacks a ligand, and the bounded null is equally consistent with the site being held by something other than IP₃. Third, the paired identities are measured against the human reference, so a module difference is partly a difference in distance from human; the column-level metrics, which are not referenced to one sequence, agree. Fourth, the pocket is measured on human IP₃R3 and carried to the other two paralogues by alignment, so a shell assignment in *ITPR1* is one transfer removed from a structure. A PI-PLC whose catalytic halves have drifted past the profiles would be scored absent; that moves a taxon into the test group and can only weaken the null, not create it.

Methods

Scope

The module comparison runs on three per-paralogue orthologue alignments of vertebrate genome-sweep gene models (the construction is described in companion paper 5): 260 *ITPR1*, 246 *ITPR2* and 262 *ITPR3* orthologues enter the paired test after the coverage rule below, which drops three, two and two. The pocket is measured in six IP₃-bound human IP₃R3 depositions (6DQN, 8TKG, 8TKF, 8TKH, 7T3P, 8TLA). The enzyme panel covers 3,527 eukaryotic reference proteomes, the 763 vertebrate sets and 2,764 others. The lineage test covers all 662 non-vertebrate reference proteomes that carry an IP₃ receptor, of which 486 enter. Prokaryotes are outside the question and were not swept.

Module definitions

Each module has a primary definition derived from measurement and a sensitivity definition taken from how the field draws the region. The primary ligand core is the minimal contiguous span containing every measured IP₃ contact, carried to each paralogue through the alignment. The primary pore module is the Pfam channel domain (PF00520 [15]) less the luminal loop, which is located geometrically as the contiguous run of channel residues whose C α lies beyond the membrane on the luminal side of the measured axial span of 6DQN. The sensitivity definitions are the published binding core, *ITPR1* residues 224–604 [3], and PF00520 as InterPro [16] draws it. Every test was run under all four combinations. Each definition must hold what it is defined to hold and none of what the other holds (all ten contacts and no pore site; both filter and both gate residues and no contact), and a failure raises rather than being written.

Ligand shells

Every residue within 15 Å of IP₃ was measured in each deposition with a minimal all-atom mmCIF reader, taking the first model and altloc A and measuring within one subunit. A C α trace puts an arginine several ångström further from a phosphate than the side chain that binds it, so all atoms are used, hydrogens included where a deposition models them; the same distances by heavy atoms alone are committed beside them (`contact_rule.tsv`). The positive control is a hard failure: all ten contacts published for 6DQN must be recovered at ≤ 4.5 Å in 6DQN or the stage stops. Residues were assigned to four shells by their median distance across the six structures (≤ 4.5 , 4.5–8, 8–11.5 and 11.5–15 Å), and residues beyond 15 Å are absent rather than pooled into an open bin. Shells were carried from IP₃R3 to the other two paralogues by alignment, and a residue that does not transfer is dropped and counted rather than assigned to a neighbour.

Paired module comparison

Each orthologue's identity to its paralogue's human reference was computed over each module's reference residues. An orthologue enters only if it resolves at least half of both modules; the excluded ones are listed with the module that lost them, because an N-terminally truncated gene model would otherwise enter as an extreme ligand-core divergence. Core-minus-pore differences were tested by Wilcoxon signed-rank and sign tests, and all tests in the study were corrected together by Benjamini–Hochberg [17]. Column-level contrasts used the sequence-weighted [18] Jensen–Shannon divergence [11] beside the composition-free share of the modal residue.

Contacts and pocket

Contact constraint was tested by permutation (200,000 iterations), resampling which positions carry the contact label against two backgrounds: the rest of the core and the rest of the pocket. The gradient was tested by Spearman correlation of distance with conservation across the 125 pocket residues. Per-site rates were taken from FEL [14] runs on the vertebrate codon alignments, reporting the share of sites called purifying at $q \leq 0.05$ per shell and comparing module β by Mann–Whitney test.

Enzyme panel

Profiles for both halves of the PI-PLC catalytic barrel (PF00387 and PF00388) were searched with hmmsearch [19] over each eukaryotic group's reference proteomes at $E \leq 10$, and the call taken by filtering the same output at $E \leq 10^{-5}$. Hits were attributed to proteomes by a measured accession-to-proteome map rather than by taxon. A proteome carries a PI-PLC only when one protein holds both halves; each half is also counted separately. The vertebrate group is the positive control for the search. Species with no reference proteome are a separate stratum and never enter the test as absences.

Lineage test and its control

Each non-vertebrate receptor was aligned pairwise with MAFFT [20] to human IP₃R3 and scored through the same module residues as the vertebrate test. Enzyme-absent receptors were compared with enzyme-carrying ones pooled, within phylum where both cells held three records, and matched: each absent record against the median of up to three present records within 0.03 pore identity. The ryanodine receptors, carrying the same pore and no IP₃ site, were scored through the same instrument as the positive control. Power was simulated at the matched design's size across a grid of true shifts and read against the shift the control produces.

Controls on the analysis itself

Forty-four constructed negative controls ran before any table was written, each built to break one rule (a core missing a contact, a pore holding the loop, a C α -only reader, a permutation scoring the whole module, an unreferenced proteome filed as an absence, a matched test that could not fire), and ten deliberate code mutations were each caught by the control responsible.

Figure legends

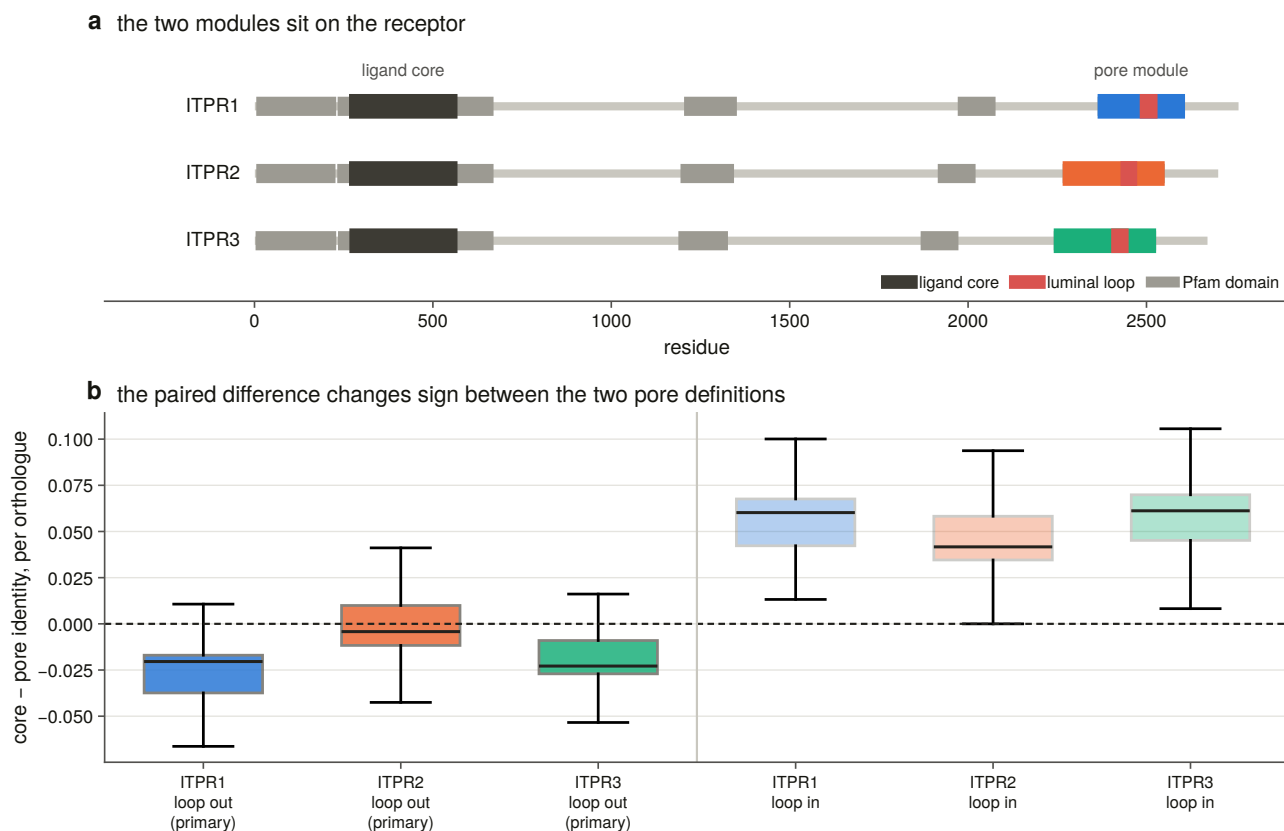


Fig. 1. The pore module is more conserved than the ligand core until the luminal loop is counted as pore, and then all three paralogues reverse. **(a)** Dark blocks mark the binding module at the N-terminus of each paralogue, coloured blocks the channel at the C-terminus, red the excluded stretch and grey the other Pfam domains. **(b)** Each box summarises the core-minus-pore identity difference, one value per orthologue, with the loop excluded (left, the primary definition) and included (right, as InterPro draws the domain); the dashed line marks zero. Both definitions share one axis because the sign of the answer turns on that single boundary.

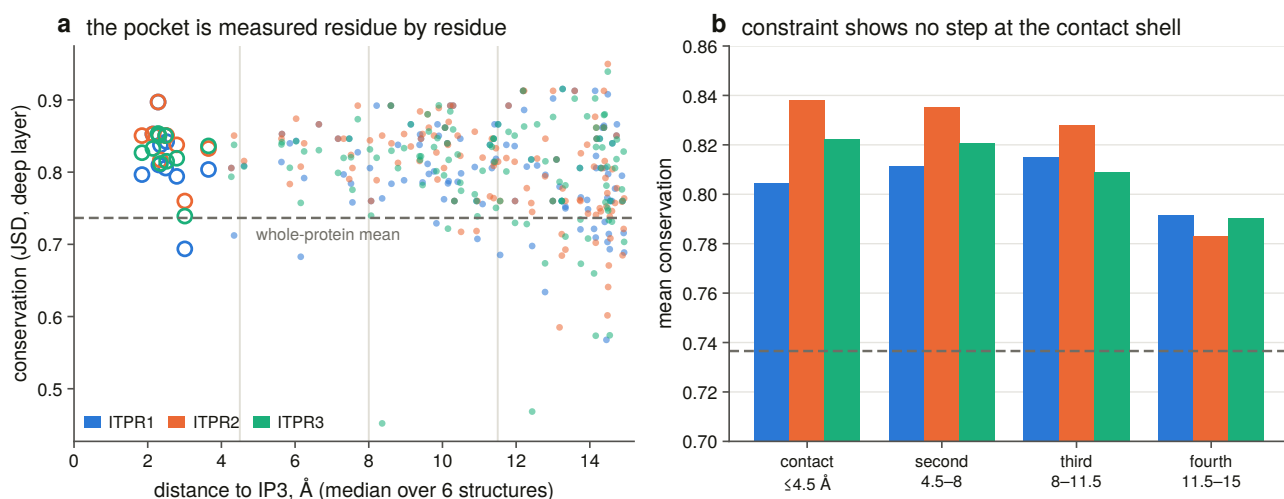


Fig. 2. Constraint declines smoothly across the ligand pocket and shows no step at the contact radius. **(a)** Each point is one pocket residue, placed at its median all-atom distance to IP₃ over six structures and coloured by paralogue; open circles are the consensus contacts, grey lines mark the shell boundaries, and the dashed line is the whole-protein mean. **(b)** Bars give the mean conservation per shell and paralogue. A contact-driven model predicts a drop beyond 4.5 Å, and none appears, which is why the constrained unit is read as the pocket.

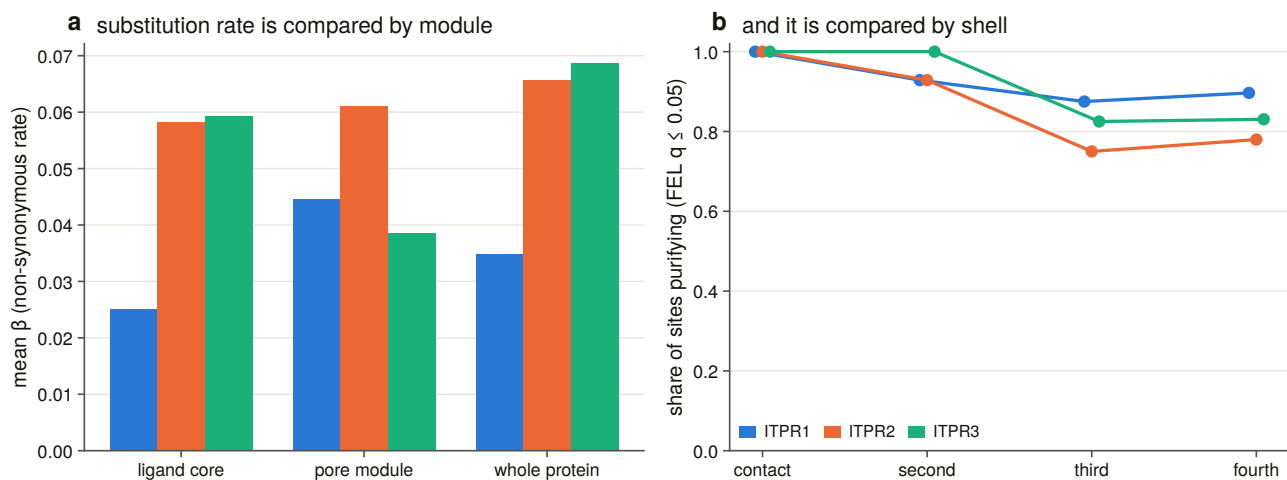


Fig. 3. Per-site substitution rates reproduce the pocket but not the module ranking. **(a)** Bars compare the mean nonsynonymous rate of the ligand core, the pore module and the whole protein in each paralogue. **(b)** Lines trace the share of sites FEL calls purifying in each shell, which falls from every contact site to a floor in the outer shells. The rate axis uses a different alignment and statistic from the identity axis, so agreement on the pocket is corroboration and disagreement on the modules is reported rather than resolved.

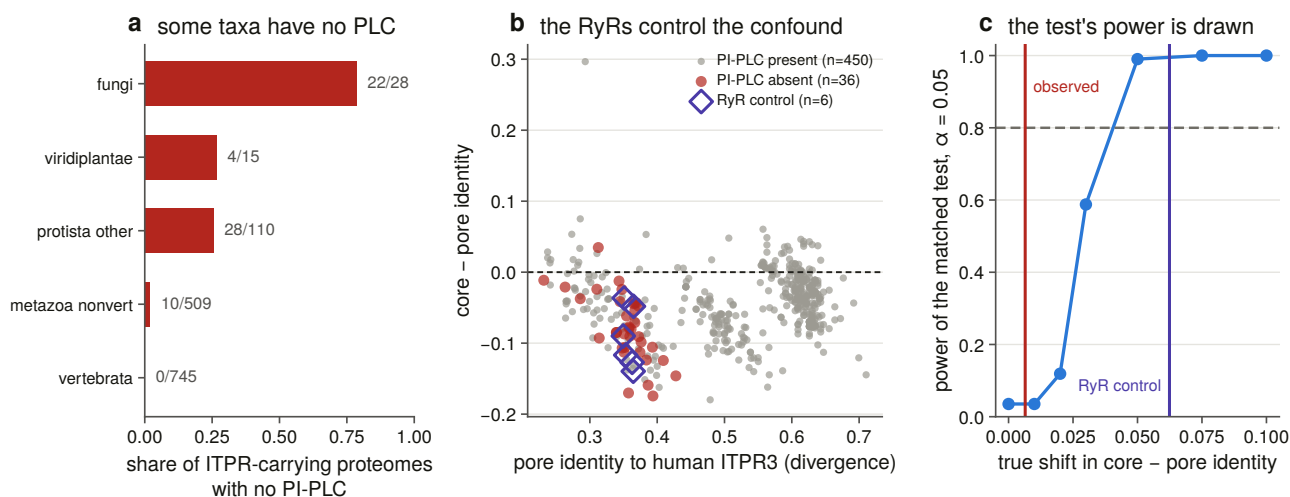
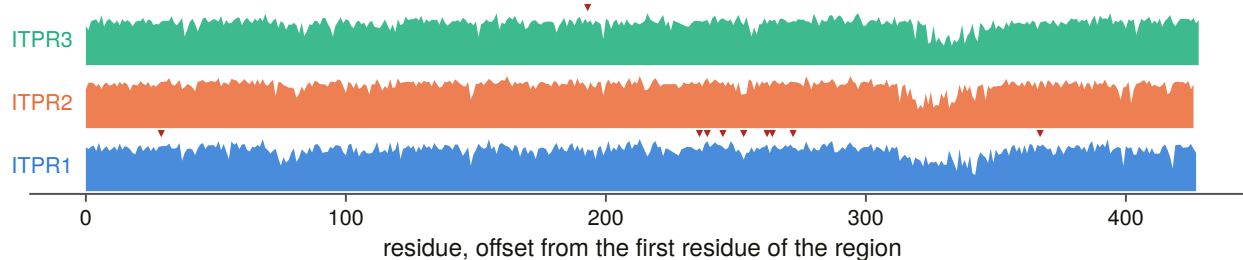


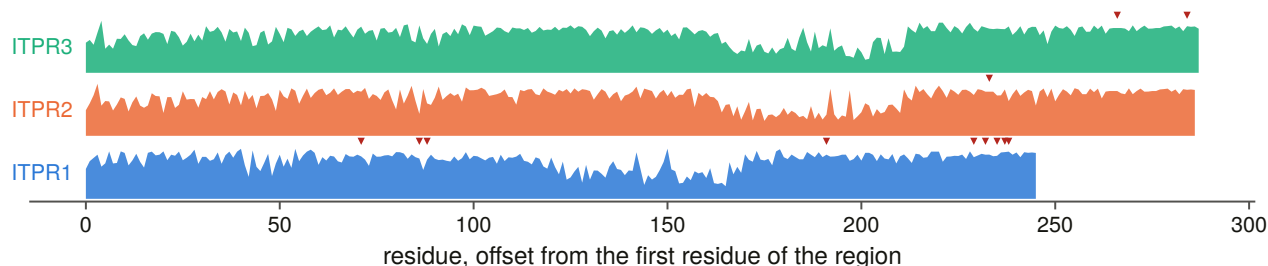
Fig. 4. Receptors in lineages without the enzyme that makes IP_3 show no relaxation of the ligand core once divergence is matched. **(a)** Bars give the share of receptor-carrying proteomes in each sweep group that lack a phosphoinositide-specific phospholipase C, with counts beside each bar; no vertebrate proteome lacks one. **(b)** Each point plots one non-vertebrate sequence's paired difference against its divergence from human IP_3R3 ; red records sit among distant sequences, which is the confound matching removes, and violet diamonds (ryanodine receptors) show where a ligand-free core falls at that distance. **(c)** The curve gives the matched test's power against the true shift, with the observed shift and the ryanodine receptor control marked, so the null is read against what the design could detect.

Extended Data

a the IP3-binding core (β -trefoil + MIR) carries deep-layer constraint; each marker is a pathogenic position



b the pore module spans channel, filter, gate and luminal loop



c every pathogenic position in the two regions is listed with the residue each paralogue carries there

		ITPR1/2/3						ITPR1/2/3			
		1	2	3				1	2	3	
ITPR1 34	β-trefoil (PF08709)	D	D	D	→N	ITPR1 2449	channel (TM)	I	I	I	→V
ITPR3 196	β-trefoil (PF08709)	A	A	A	→T	ITPR1 2451	channel (TM)	L	L	L	→P
ITPR1 241	MIR	R	R	R	→K	ITPR1 2592	channel (TM)	L	L	L	→P
ITPR1 244	MIR	H	H	H	→P	ITPR1 2600	channel (TM)	T	T	T	→P
ITPR1 250	MIR	F	F	F	→L	ITPR1 2601	channel (TM)	F	F	F	→L
ITPR1 258	MIR	K	K	G	→M	ITPR3 2506	channel (TM)	I	I	I	→N
ITPR1 267	MIR	T	T	T	→R	ITPR3 2524	channel (TM)	R	R	R	→H
ITPR1 269	MIR	R	R	R	→L	ITPR1 2554	filter	G	G	G	→R
ITPR1 277	MIR	S	S	S	→I	ITPR2 2498	filter	G	G	G	→S
ITPR1 372	MIR	P	A	P	→L	ITPR1 2595	gate	G	G	G	→A
ITPR1 2434	channel (TM)	T	T	T	→I	ITPR1 2598	gate	I	I	I	→T

a red letter marks a residue the other paralogue does not share; every letter was checked against that paralogue's own per-residue table

Supplementary Fig. 1. Both functional modules are drawn at residue resolution in all three paralogues, so every boundary can be checked by eye. **(a)** The track shows deep-layer constraint along the β -trefoil and MIR region of each paralogue in its own numbering, with every pathogenic position marked. **(b)** The channel region is drawn the same way, with the selectivity filter, the gate and the geometrically located luminal loop beside the pore module rather than inside it. **(c)** The table prints the residue each paralogue carries at every pathogenic position in the two regions, and a red letter marks a paralogue that differs; each letter was checked against that paralogue's own per-residue table before it was drawn.

Data availability

Every table this paper's numbers and figures are drawn from is committed under `results/ligand_site/` in the project repository, with the SHA-256 of each recorded in `ligand_site_stats.json` and in this paper's deposit manifest. Bulk inputs (reference proteomes, structure files and per-genome sweep output) are public and are listed with the command that regenerates each in `deposit_notes.md`.

Code availability

The analysis is `scripts/s22_run.py` with its modules and negative controls; this paper is built by `scripts/s26_assemble.py --paper ligand`, which re-verifies every load-bearing number against its source table.

Author contributions

George Dickinson directed the project, set its questions and reviewed its results. The analysis code and the text were written by Claude (Anthropic) under his direction; Chapter 16 of the project thesis describes how.

Competing interests

The author declares no competing interests.

Companion papers

5. George Dickinson. Purifying selection on the IP₃ receptor is strongest at the gate and filter and weakest in a luminal loop beside them. *In preparation* (paper 5 of 6 in this series; source in `papers/constraint/`).

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Citations are rendered from `results/s0_baseline/references.tsv`, numbered in order of first appearance.

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